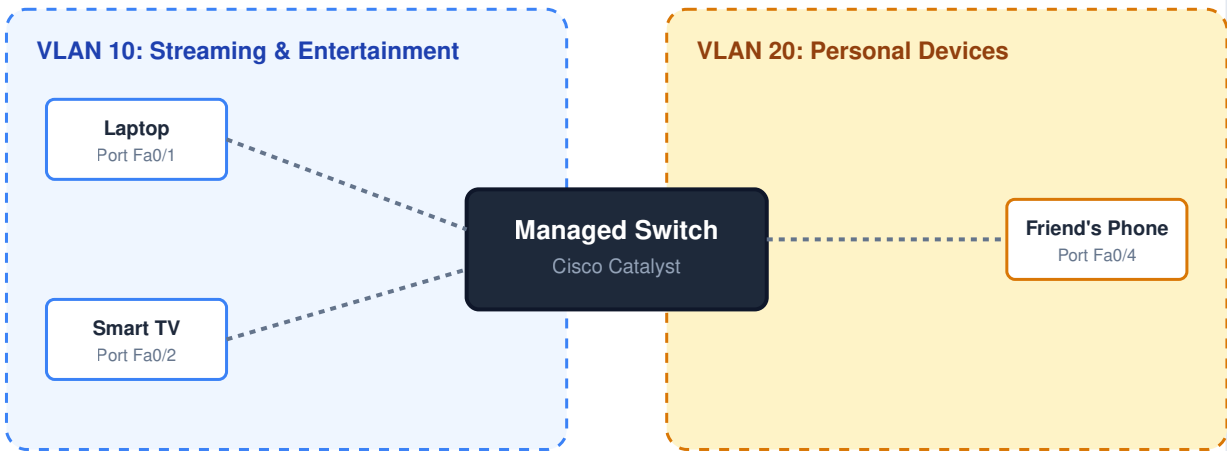


VLAN Configuration & Network Segmentation Guide

Comprehensive Solutions for Network Topology, Switch Configuration, Dynamic Port Migration, and Bulk Automation

Task 1: Network Topology & VLAN Architecture

Below is a visual network diagram illustrating how multiple endpoint devices connect to a central managed switch, segmented into distinct Virtual Local Area Networks (VLANs) for isolation and optimized traffic management.



Architecture Breakdown:

- **VLAN 10 (Streaming & Entertainment):** Houses high-bandwidth or media-focused devices such as the user's laptop (Port Fa0/1) and Smart TV (Port Fa0/2).
- **VLAN 20 (Personal Devices):** Isolates visitor/personal mobile devices like the friend's phone (Port Fa0/4) to restrict direct lateral movement into primary network shares.

Task 2: Initial Switch Configuration & VLAN Creation

In a network simulator (such as Cisco Packet Tracer or NetSim), switches come with a single default VLAN (VLAN 1) where all ports reside initially. For proper segmentation, we create specific VLANs and assign access ports accordingly.

Configuration Script

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-Core
SW-Core(config)# vlan 10
SW-Core(config-vlan)# name STREAMING
SW-Core(config-vlan)# exit
SW-Core(config)# vlan 20
SW-Core(config-vlan)# name PERSONAL
SW-Core(config-vlan)# exit

! Assigning ports to VLAN 10 (PCs 1, 2, 3)
SW-Core(config)# interface range fastEthernet 0/1 - 3
SW-Core(config-if-range)# switchport mode access
SW-Core(config-if-range)# switchport access vlan 10
SW-Core(config-if-range)# exit

! Assigning ports to VLAN 20 (PCs 4, 5)
SW-Core(config)# interface range fastEthernet 0/4 - 5
SW-Core(config-if-range)# switchport mode access
SW-Core(config-if-range)# switchport access vlan 20
SW-Core(config-if-range)# end
SW-Core# write memory
```

Verification & Isolation Testing

To verify that segmentation is working correctly, we test connectivity between PCs across different VLANs:

- **PC1 (VLAN 10 - IP: 192.168.10.11) pinging PC2 (VLAN 10 - IP: 192.168.10.12):**

Success (Reply received)

- **PC1 (VLAN 10 - IP: 192.168.10.11) pinging PC4 (VLAN 20 - IP: 192.168.20.14):**

Failed (Request timed out)

Because VLANs create separate Layer 2 broadcast domains, devices in VLAN 10 cannot communicate directly with devices in VLAN 20 without an intervening Layer 3 router or multilayer switch configured with inter-VLAN routing (Router-on-a-Stick or Switched Virtual Interfaces).

Task 3: Moving a Device Between VLANs

Network requirements frequently change, requiring device ports to be reassigned to different VLANs. In this scenario, we move **PC3** from **VLAN 10** (Port FastEthernet 0/3) to **VLAN 20**.

1. Before Configuration State

Port Fa0/3 is assigned to VLAN 10, and PC3 has IP address 192.168.10.13.

```
SW-Core# show running-config interface fastEthernet 0/3
Building configuration...
Current configuration:
!
interface FastEthernet0/3
  switchport mode access
  switchport access vlan 10
  no shutdown
end
```

2. Configuration Changes

To reassign the port to VLAN 20 and update the PC's IP configuration to match the new subnet (192.168.20.0/24):

```
SW-Core# configure terminal
SW-Core(config)# interface fastEthernet 0/3
SW-Core(config-if)# switchport access vlan 20
SW-Core(config-if)# end
SW-Core# write memory
```

3. After Configuration State & Verification

```
SW-Core# show running-config interface fastEthernet 0/3
Building configuration...
Current configuration:
!
interface FastEthernet0/3
  switchport mode access
  switchport access vlan 20
  no shutdown
end
```

Connectivity Test Results:

- Ping from PC3 (now on VLAN 20) to PC1 (VLAN 10): **Failed (Isolated across VLANs)**
- Ping from PC3 (VLAN 20) to PC4 (VLAN 20): **Success (Communicating within VLAN 20)**

Task 4: Automating Bulk Port Assignment to VLAN 30

When provisioning multiple ports for a specific department or function (e.g., 10 ports for VLAN 30), configuring each interface individually is inefficient and prone to human error. Cisco IOS supports the `interface range` command to execute commands across multiple interfaces simultaneously.

Automation Script / Command List

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 30
Switch(config-vlan)# name OPERATIONS
Switch(config-vlan)# exit

! Bulk assignment of 10 consecutive ports (FastEthernet 0/11 through 0/20) to
VLAN 30
Switch(config)# interface range fastEthernet 0/11 - 20
Switch(config-if-range)# description OPERATIONS_ACCESS_PORTS
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# spanning-tree portfast
Switch(config-if-range)# end
Switch# write memory
```

Verification Command:

```
Switch# show vlan brief

VLAN Name                Status    Ports
----
1    default                active    Fa0/6, Fa0/7, Fa0/8, Fa0/9,
Fa0/10, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   STREAMING               active    Fa0/1, Fa0/2
20   PERSONAL                active    Fa0/4, Fa0/5
30   OPERATIONS              active    Fa0/11, Fa0/12, Fa0/13, Fa0/14,
Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20
1005 default/fddi          active
...
```

Using the `interface range` command eliminates repetitive typing, ensures consistent policy enforcement across all ports, and drastically reduces configuration time.